

CONTINUATION

CONTINUATION — Helix Proxy: VPN-Aware Dynamic Routing Extension

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:** Active — control-plane + config-plane landed and unit/integration/parse-proven; the LIVE dynamic-stack data-plane proof is owed to P10. **Branch:** feature/vpn-aware-dynamic-routing **Spec:** docs/superpowers/specs/2026-06-30-vpn-aware-proxy-extension-design.md (Rev 4) **Plan:** docs/superpowers/plans/2026-06-30-vpn-aware-proxy-extension-plan.md (Rev 1) **Authority:** Inherits the Helix Constitution submodule (constitution/Constitution.md) per §11.4.35.

§12.10 live-state resume file. Read this first, then `git fetch --all --prune` and re-read `git log --oneline main..HEAD`. Any agent must be able to resume exactly where the last session left off from this single file.

1. Current PHASE

Control-plane / config-plane construction phase — landed: P0-P5a + P7.1/7.2/7.3. 14 commits on feature/vpn-aware-dynamic-routing ahead of main. The Go control-plane logic (stores, health-publisher, acl-helper, config-compiler) and the DNS/observability/security config-plans are written and proven at the unit / integration / config-parse layer. **No dynamic-mode container has been booted and no live data-plane evidence exists yet** — that is the entire job of P10 (the usability proof). In-flight next: **P5b** (breaker/failover), **P9** (full test matrix), **P11** (docs export — this file is part of it). **P10 is the critical unblocker** for any “it works for the end user” claim.

2. The 14 landed commits (`git log --oneline main..HEAD`, newest first)

#	Commit	Phase	Summary
14	1833c8f	P7.3	per-user Squid auth + rootless Podman-secret loader + kill-switch design (no secrets)
13	603e039	P5a	acl-helper — Squid external_acl OK/ERR from Redis, fail-closed (stdlib)
12	e6e336f	P7.1	per-tunnel DoH/DoT (dnsproxy) config plan + DNS-leak test design
11	04526dd	P4	config-compiler — render Squid/Dante/PAC from PG + seed route keys (parse-verified)
10	833fb9e	P7.2	Prometheus scrape + Grafana dashboard config plan (promtool-validated)
9	11106a4	P3	vpn-health-publisher (cmd/healthd + internal/vpn) — data-plane health, fail-closed, TDD
8	b66d172	P4	Squid 6.13 + Dante dynamic-mode templates (additive, parse-verified) + spec reconcile
7	fbfe9ed	P1	docs(spec): mark §20 gaps G1-G4 RESOLVED with spike decisions; bump to Revision 2
6	e19e0ed	P2	store (pgx) + redis (go-redis) clients — fail-closed, TDD, real PG/Redis
5	6409cb9	P1	docs(research): resolve spec §20 gaps G1-G4 with captured-evidence spikes
4	6802798	P1/E	docs(audit): §11.4.138 forensic bluff-audit of 4 existing test scripts (8 bluffs)

#	Commit	Phase	Summary
3	9ac1b4a	P0	docs(dynamic-routing): DYNAMIC_ROUTING.md + 2 mermaid diagrams
2	6251007	P0	chore(submodules): incorporate containers, helix_qa, challenges, docs_chain (SSH, no- force)
1	5f917a7	P0	P0 scaffold — data model, evidence harness, Go skeleton, governance carriers

3. PROVEN-NOW vs OWED-TO-P10 (honest §11.4.6)

PROVEN-NOW (control-plane / config-plane / spike facts — captured)

- **Spec §20 gaps G1-G4 resolved** with transient-spike captured evidence (docs/research/mvp/findings/F_spikes_G1-G4.md, run-id qa-results/spikes/20260630_205029_g1g4/): G2 ubuntu/squid:latest = Squid **6.13** (not v8), §8 directive set squid -k parse exit 0; G4 gluetun **v3.40 (=v3.40.4)** control-API :8000 answers 200, issue #3060 confirmed; G1 kernel-WG interface **creatable rootless** with --cap-add NET_ADMIN; G3 Dante **SIGHUP preserves an active SOCKS session** (20/20 chunks, curl exit 0, /proc/net/tcp ESTABLISHED proof).
- **P2 stores** (pgx + go-redis) — fail-closed, TDD, exercised against **real PG / Redis**.
- **P3 vpn-health-publisher** (cmd/healthd + internal/vpn) — data-plane health poll → Redis state, fail-closed, TDD.
- **P4 config-compiler + templates** — Squid 6.13 (%>ha{Host}) + Dante (concatenation, no include) render from PG; **squid -k parse exit 0**; PAC + route-key seeding parse-verified.
- **P5a acl-helper** — Squid external_acl OK/ERR from Redis, **fail-closed**, stdlib-only.
- **P7.2 observability** config plan — **promtool-validated** Prometheus scrape + Grafana dashboard.
- **P7.1 DNS / P7.3 security** — config plans only (DoH/DoT per-tunnel; per-user auth + Podman-secret loader + in-netns kill-switch). Design + parse layer.
- **Existing-test bluff audit** (Stream E) — 8 bluffs across 4 scripts catalogued (§11.4.138), guards owed to P8.

OWED-TO-P10 (the LIVE dynamic-stack data-plane proof — NOT yet captured)

The dynamic compose profile (postgres + redis + control-plane + gluetun(s) + squid+helper + dante) has **never been booted**; docs/DYNAMIC_ROUTING.md is explicitly DESIGN-only. The following remain **unproven live** and are the usability proof: -
vpn_real_egress — egress IP via proxy == tunnel exit && != host IP + wg transfer Δ (200 OK is not routing). - graceful_503 — tunnel down → branded 503 with **Squid PID unchanged** → up → 200, live. - no_leak / killswitch — drop tunnel → **zero** target packets on the real uplink (tcpdump) + DNS only via the intended resolver. - per-user **407 auth challenge** live; **secret injection leak-free** at runtime. - **G1 residual** — full rootless kernel-WG *operation* (handshake + routing + throughput), only interface *creation* was spiked (§20 G1). - **G3 residual / P9** — concurrent / repeated SIGHUP + **route-change-mid-session** SOCKS path behaviour (§20 G3). - circuit-breaker open → failover to next up tier (P5b not yet landed).

4. Remaining phases

Phase	Scope	State
P5b	per-target circuit breaker + tunnel tier-failover (sony/gobreaker/v2)	next, in flight
P6	control-API + admin UI (templ+htmx+SSE, OpenDesign §11.4.162, host-rendered pixel proof §11.4.170)	pending
P8	fix the existing-test bluffs (Stream E — the 8 audited bluffs → real probes / honest SKIP + §11.4.135 guards)	pending
P9	full test matrix + Challenges + HelixQA (all §11.4.169 types; G3 route-change-mid-session live test)	in flight
P10	live dynamic-mode boot + captured	critical unblocker

Phase	Scope	State
	data-plane evidence = the usability proof	
P11	docs sync + HTML/PDF (+DOCX where mandated) exports (this CONTINUATION + .remember are part of it)	in flight
P12	whole-branch review + full retest + merge to main + prefixed release tag	last

5. Binding constraints (non-negotiable)

- **Anti-bluff §11.4** — every PASS carries positive captured **data-plane** evidence; control-plane/config-parse green is necessary, never sufficient; the end-user-usability bar is met only at P10.
- **No force-push §11.4.113** — merge onto latest main, fast-forward only; force-push is forbidden with no exception.
- **Rootless Podman §11.4.161** — all containers rootless; no Docker-rootful, no sudo, no root escalation; orchestrate via the containers submodule (§11.4.76), build on the remote host (§11.4.173).
- **Secrets-as-names-only §11.4.10** — VPN creds / proxy-auth / mTLS keys via Podman secrets / file refs; **never** plaintext in git; .env.example documents refs only.
- **Operator-safe §11.4.174** — do **NOT** touch the operator's pre-existing resources: the host wg0-mullvad (UP kernel-WG) interface and any lava-* containers (e.g. lava-postgres-thinker) are off-limits; verify process/resource ownership before acting; block-don't-break on shared-host contention.
- **Host safety §12** — ≤60% memory (§12.6); no host power-state commands (CONST-033); pull images sequentially; --rm diagnostics; df first.

6. Resume now (next actionable)

1. git fetch --all --prune on feature/vpn-aware-dynamic-routing; confirm HEAD 1833c8f (or integrate any newer foreign commit per §11.4.71, no force).
2. In flight in parallel (§11.4.103): **P5b** breaker/failover, **P9** test matrix, **P11** docs export. None of these is the usability proof.

3. **Unblock P10** as the priority — it is the only phase that converts the landed control-plane into captured end-user-usable evidence; until it lands, no “works for users” claim may be made (§11.4 / §11.4.108).
4. Every change: TDD reproduce-first (§11.4.43/§11.4.115), all warranted test types (§11.4.169), paired §1.1 mutation, independent review → iterate-to-GO (§11.4.142/§11.4.125/§11.4.134), docs in sync (§11.4.60/§11.4.65/§11.4.106).